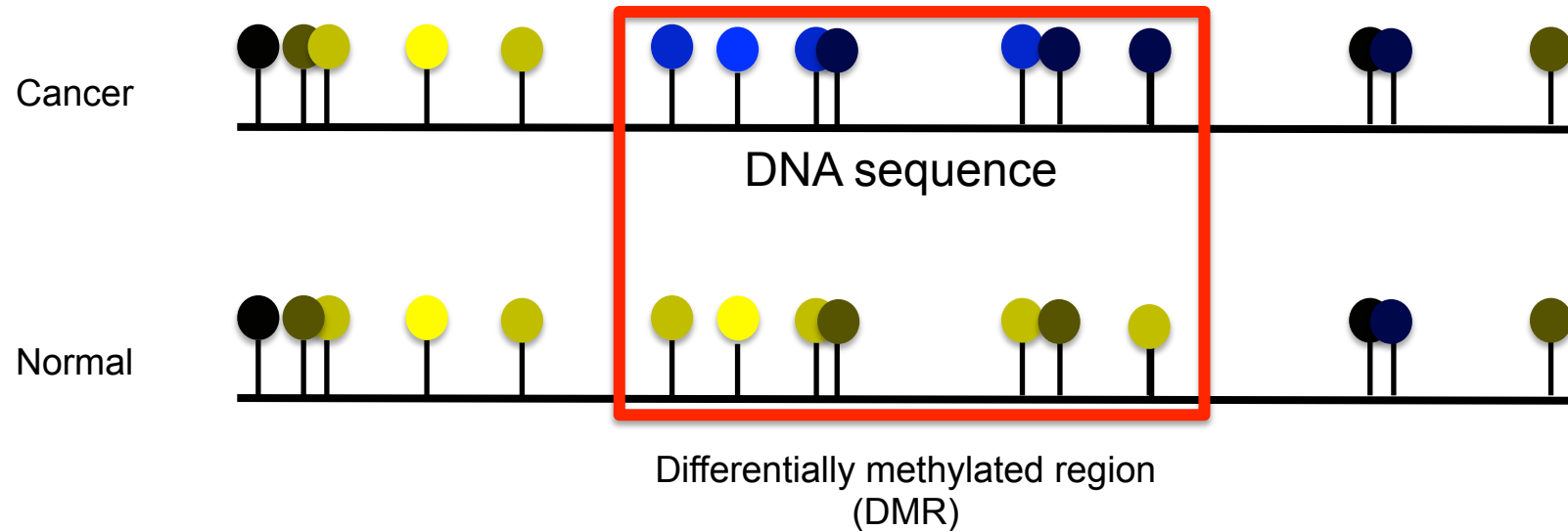


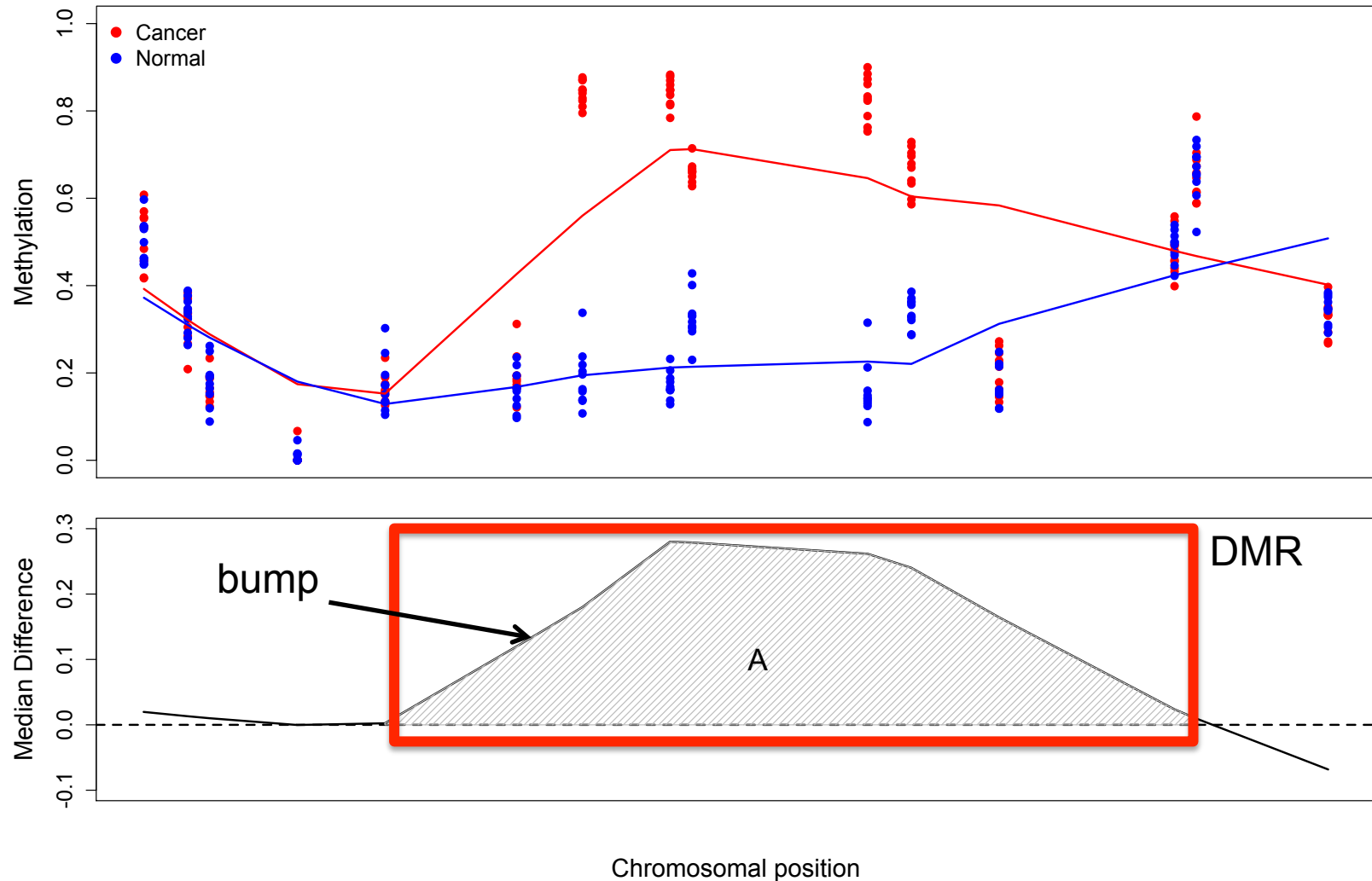
Challenges presented by local dependence in DNA methylation

- Identification of differentially methylated CpGs
- Many multiple testing procedures assume independent tests
- Problematic when test statistics are highly correlated:
 - May inflate the variance of the number of false discoveries (Owen, 2005) and FDR estimators (Qiu & Yakovlev, 2006)

Identifying Differentially Methylated Regions (DMRs)



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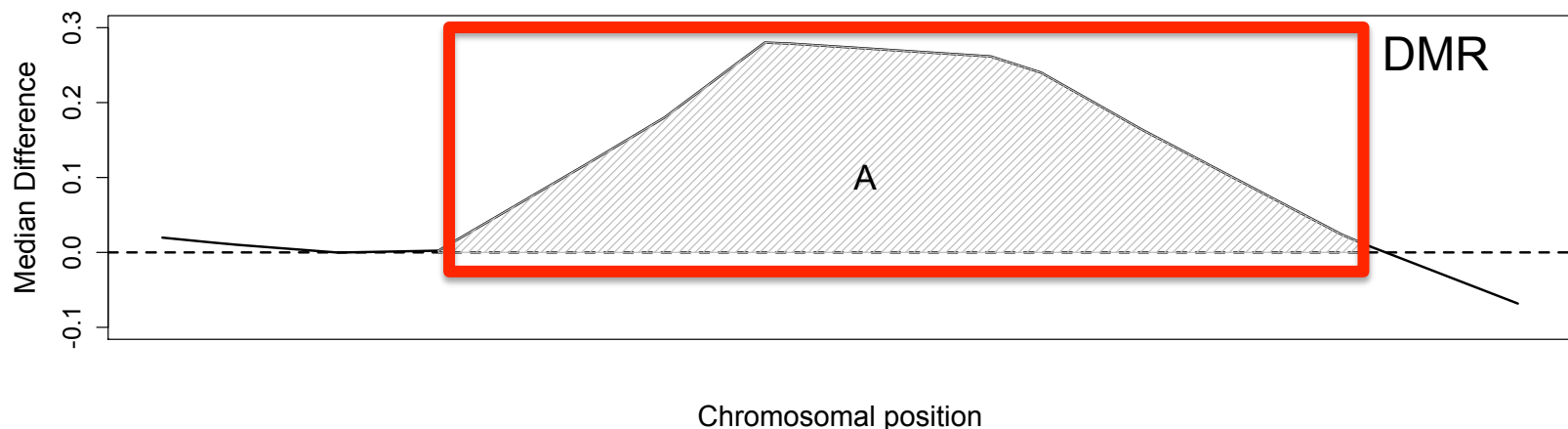
- How do we define a region?
 - Need to know where to look
- Identifying DMRs?
- Testing the statistical significance DMRs?
- Controlling for multiplicity?
- Jaffe et al. (2012) formalize these concepts
 - “Bump hunting to identify differentially methylated regions in epigenetic epidemiology studies”

A Brief History of Bump-Hunting

- Bump hunting or “peak detection”
- Finding transcription factor binding sites with (ChIP-chip) data
- Key differences between DNA methylation and ChIP-chip
 - DNA methylation data is typically much larger
 - Observed correlation structure
 - ChIP-chip, peaks expected to be triangular shapes ranging several hundred base pairs
 - Expect a “larger number of bumps” in DNA methylation data

Bump-Hunting in the context of DNA methylation data

- Identify associations at the region level as opposed to the single CpG level
- Amounts to finding spatial intervals in which an estimated function is different from 0
 - Average difference between two conditions
 - Correlation with a continuous trait



Step 1: CpG-by-CpG analysis

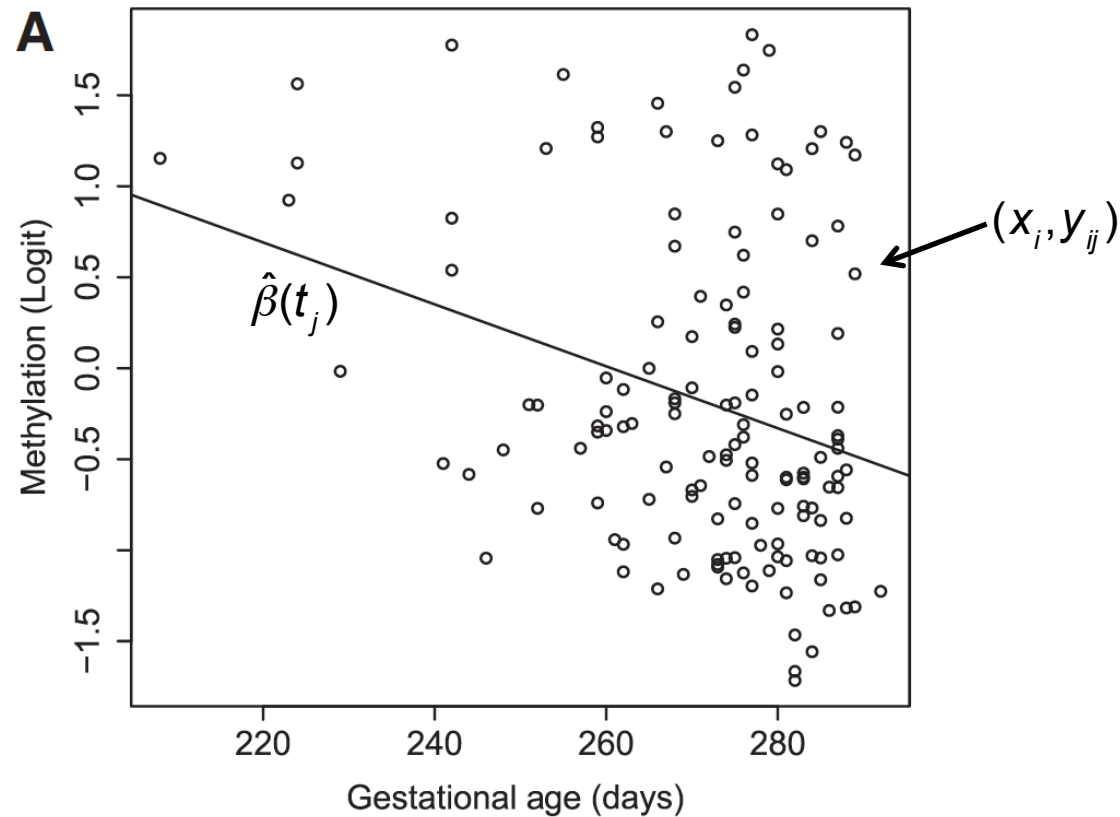
- Y_{ij} : represents methylation for subject i CpG j
- Begin by fitting models of the form:

$$Y_{ij} = \alpha(t_j) + \beta(t_j)X_i + \sum_{p=1}^P \gamma_p(t_j)Z_{ip} + e_{ij}$$

- t_j location on the genome of j^{th} CpG
- X_i represents some phenotype/exposure
- Z_{ip} potential confounders
- $\text{var}(e_{ij}) = \sigma_j^2$

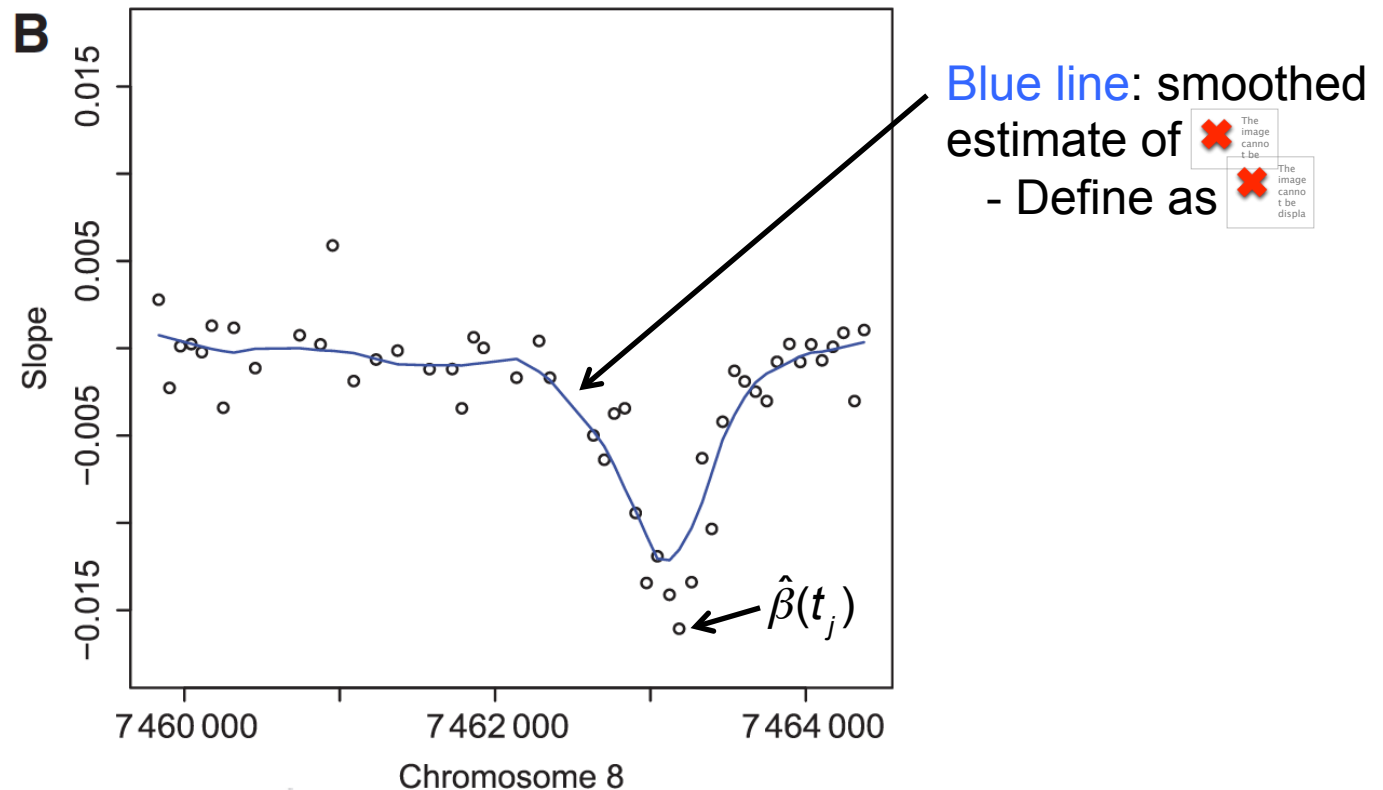
Step 1: CpG-by-CpG analysis

Plot of methylation (CpG j) by gestational age (days)



Jaffe et al. (2012)

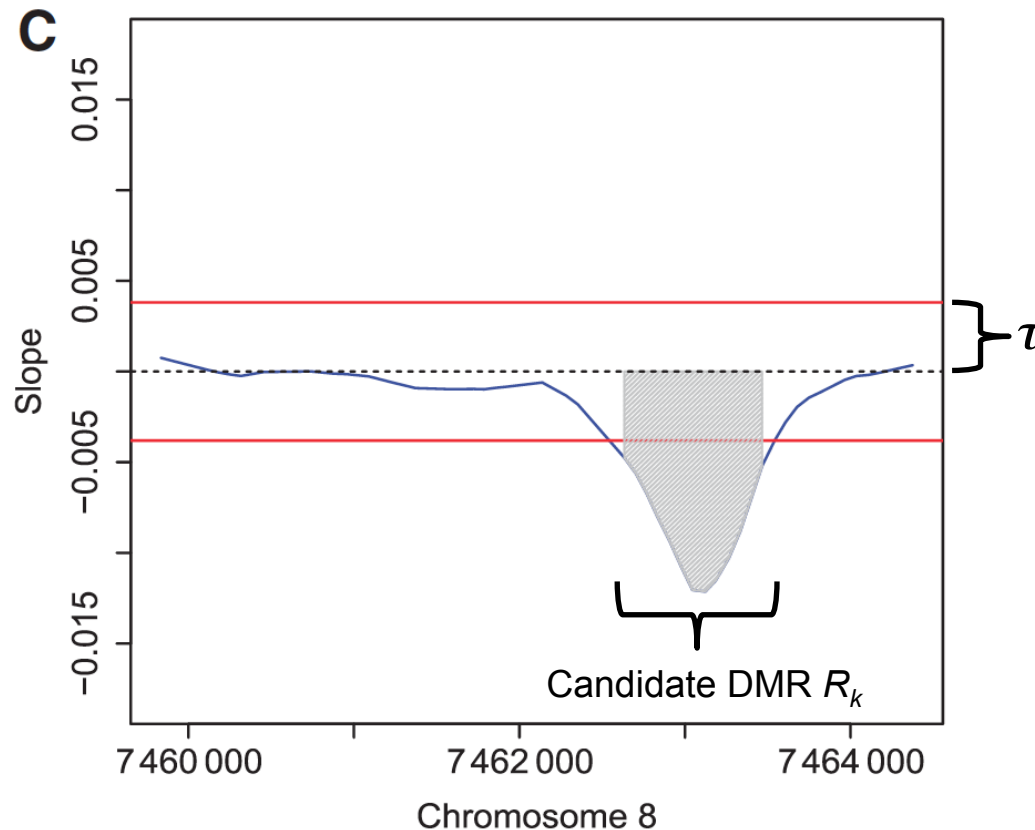
Step 2: Smoothing & Defining regions



- Contiguous intervals R_k , $k = 1, 2, \dots, K$ for which $\beta(t) \neq 0$ for all $t \in R_k$

Step 3: Identifying candidate DMRs

- Candidate DMRs are defined as regions consisting of contiguous runs of measurements where $|\tilde{\beta}(t)| > \tau$

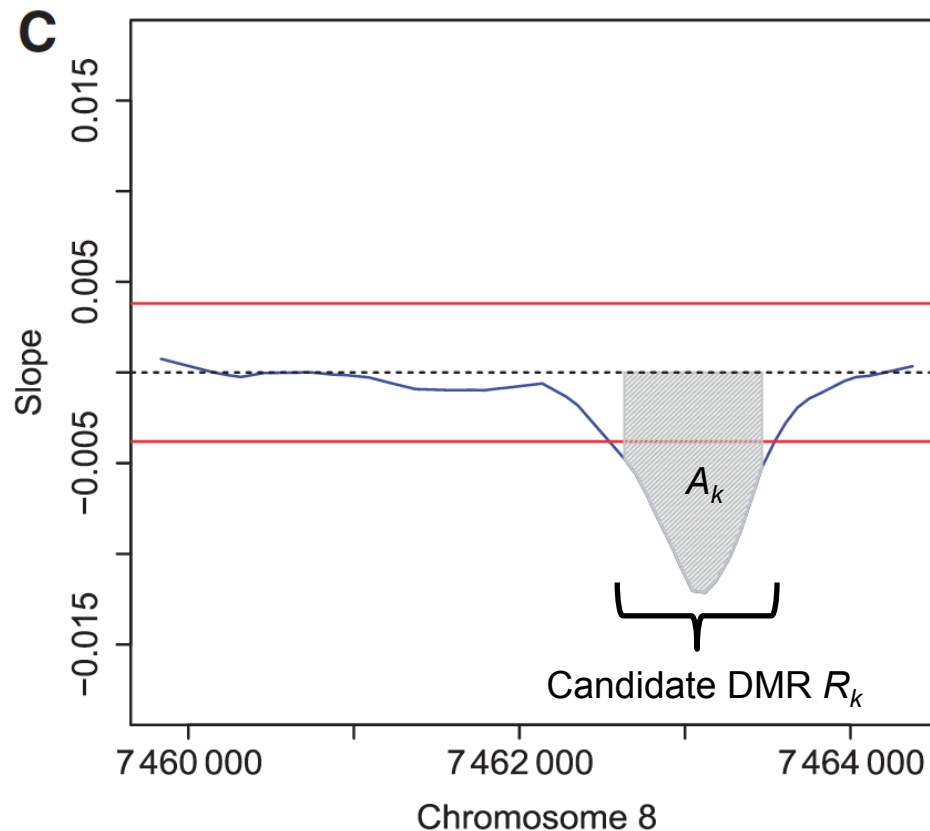


Jaffe et al. (2012)

Step 4: Assessing statistical significance candidate DMRs

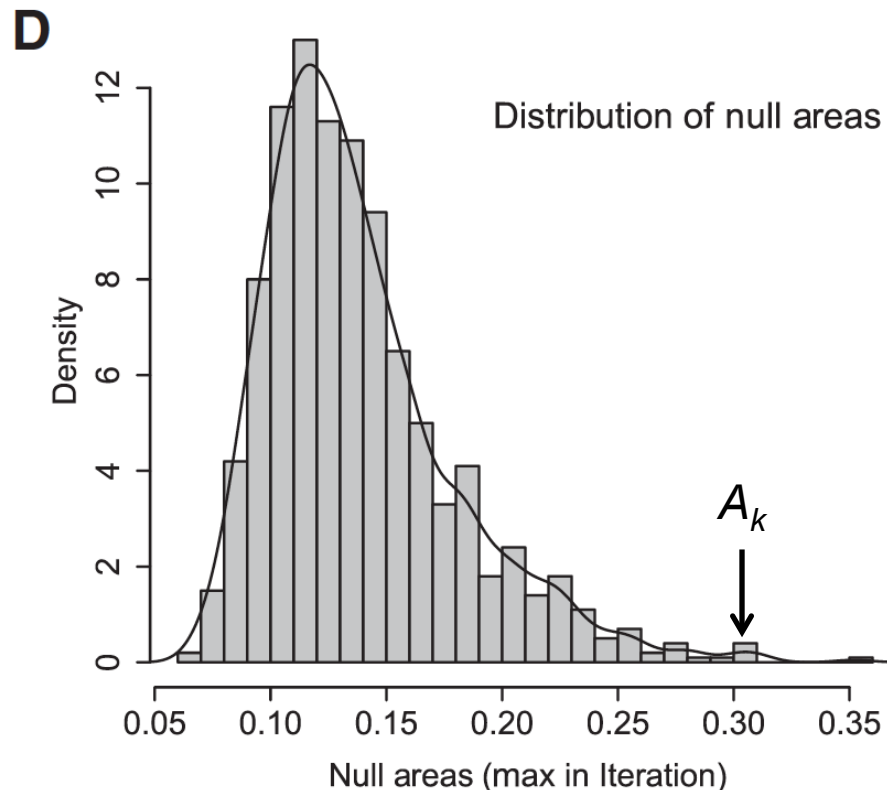
- Strength of evidence for each region is summarized by its area:

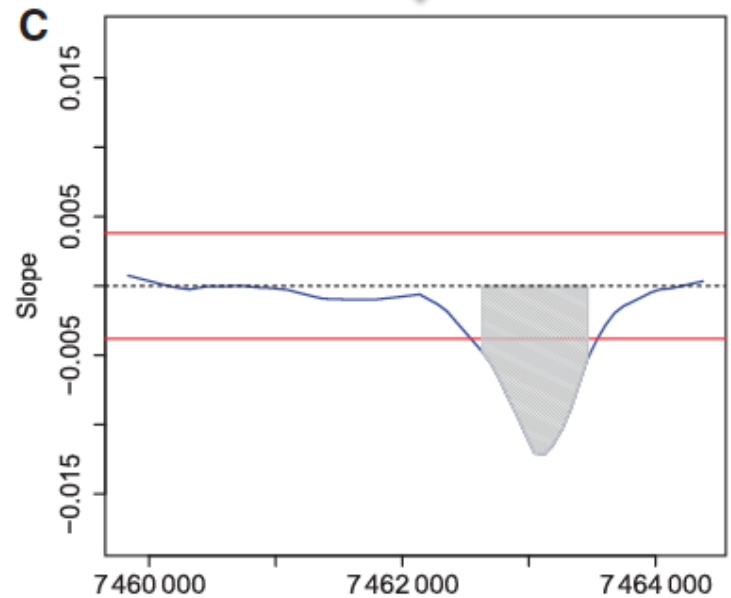
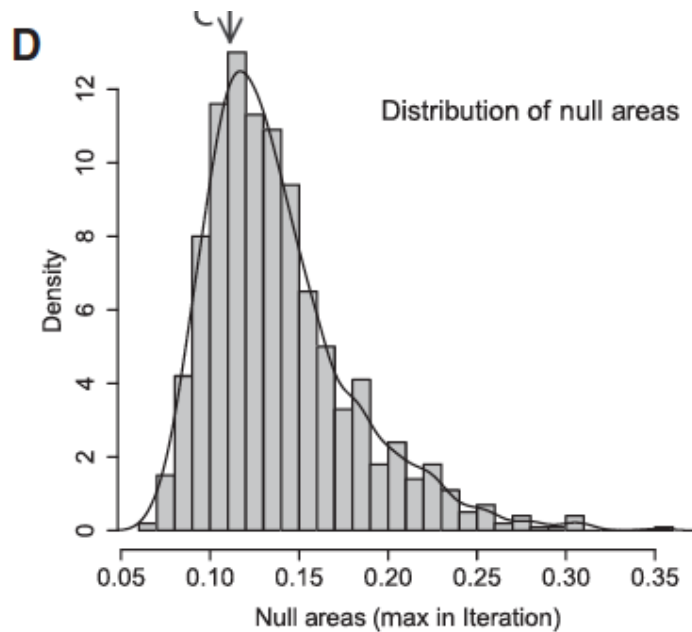
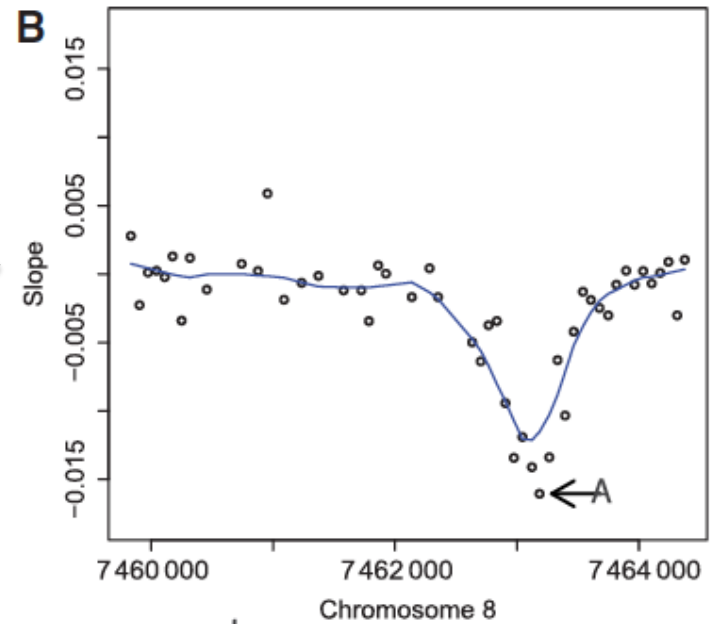
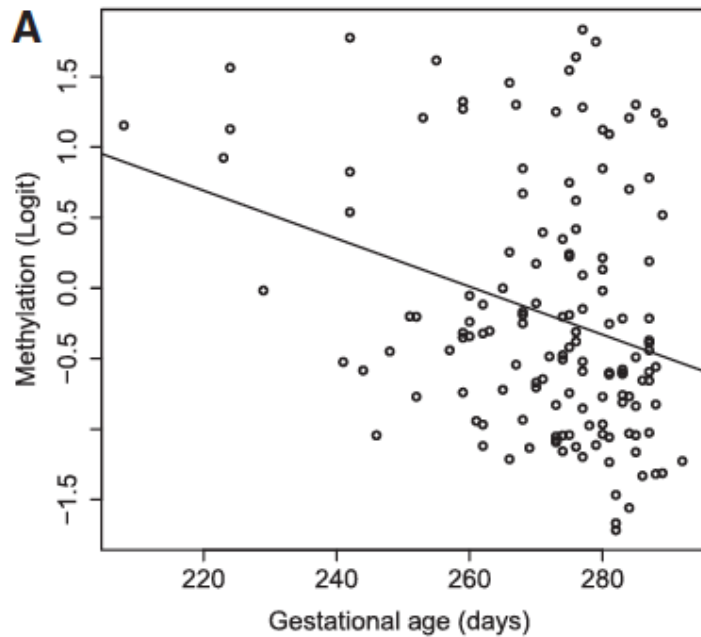
$$A_k = \sum_{j \in R_k} |\tilde{\beta}(t_j)|$$



Step 4: Assessing statistical significance candidate DMRs

- Permutation test:
 - Permute the phenotype/exposure x_i
 - Compute A_k^* , $k = 1, 2, \dots, K^*$
 - Repeat 1000 times: $A_{k,b}^*$, $k = 1, 2, \dots, K_b^*$





Implementation of the bump-hunting procedure

- “dmrFinder” function in the BioC package charm

```
source("http://bioconductor.org/biocLite.R")  
biocLite("charm")
```

- Okay for 450K data, but not for 27K data
 - 27K array is too sparse
- Currently exploring more reasonable ways to compute the p-values for each of the identified bumps